

PROCUREMENT AND BUSINESS ANALYSIS

Hardware, Model & Serving Strategy

Nine options across three budget tiers, priced in dirhams from UAE sellers, sized against a stated workload and judged on two baselines: what the tokens would cost from an API, and what the same capability would cost to rent.

Prepared for	Amro - syslab-server project
Compiled	4 September 2026 · Dubai, UAE
Currency	AED throughout, at UAE seller prices including VAT. USD appears only where the vendor bills in it, converted at the 3.6725 peg
Sizing target	10–20 client organisations, with a stated upgrade path
Workload	Tool-calling document agent, voice chat, vision and image reading, and graph generation
Budget ceiling	AED 80,000
Inputs	docs/hardware_report_data.json — every figure carries a named seller, a source URL and a capture date

The question this report answers

There is no production server yet. The choice is to rent capacity or to build a rig, and it is genuinely open: at the design point, renting internationally costs about the same per month as owning, while renting inside the UAE costs roughly four times as much. **So the decision is not really about hardware. It is about whether client documents are allowed to leave the country** — and this report puts a monthly figure on that answer rather than assuming one.

Prices move fast in the current memory shortage. Every figure here was captured at the compile date from a named UAE seller and can be re-checked, or the report regenerated, from the data file.

Executive summary

What to do, what it costs in dirhams, and the one answer that changes it.

Four findings

1. The workload needs 48 GB, not 96. Fifteen client organisations at four active users each generate 2,880 agent turns a day, but office traffic is bursty and short. The busiest hour carries 576 of them, and at 18 seconds of service per turn that is only 2.9 requests in flight on average. Queueing theory puts **6 concurrent slots** at under a 10% chance of waiting, and 8 covers twenty organisations. Weights, that much KV cache and vLLM's own overhead come to **47 GB** at an FP16 cache, 42 GB at FP8. Organisations are not concurrency, and confusing the two is how a 96 GB card gets bought for a 48 GB job.

2. Data residency decides rent versus buy, and it is worth about AED 2,698 a month. Renting a capable card internationally costs AED 1,598 a month all in. Renting one inside the UAE costs AED 4,067, because there is no in-region rental of a 48 GB consumer card at all — the cheapest UAE-hosted option is a datacentre H100 at four times the price for hardware far larger than this needs. Owning sits between them at AED 1,369. So if documents may leave the country, renting is competitive and reversible. **If they may not, owning is decisively cheaper and the only question left is which rig.**

3. The UAE second-hand market is the cheapest route to the design point, by a wide margin. Two used RTX 3090s reach 48 GB for about AED 4,800 of graphics card; the complete build is AED 15,569 delivered. The same 48 GB in used Ada costs AED 29,769. What the extra buys is an FP8 path that halves the KV cache, turning 48 GB from comfortable at fifteen organisations into comfortable at twenty — worth AED 14,200 only if that case is firm.

4. The free change is worth more than any of the paid ones. Ollama does not do continuous batching — it serves one request at a time. Until `app/llm.py` speaks to a batching server, **JOB_WORKERS** rises above 1 and uvicorn runs more than one process, a bigger card buys very little. That work is roughly a day, it costs nothing, and it is a precondition for every option below, rented or owned.

The recommendation, in order

	Step	Cost	Why
1	Settle the residency question	free	Ask the client, in writing, whether their documents may be processed outside the UAE. Every number in this report bends around that answer and nothing should be bought before it exists.
2	Land the vLLM seam	~1 day	Port <code>app/llm.py</code> to the OpenAI-compatible shape and lift the two other ceilings. Free, reversible, and required whether the endpoint is rented or owned.
3	Rent for a month and measure	~AED 1,692	A 48 GB card on the international line, running the real stack on Ubuntu. It replaces the modelled concurrency figure with a measured one and rehearses the Linux move before any card is bought.
4	Buy L2 — two used RTX 3090	AED 15,569	If residency is required, or if the measurement shows sustained load. Inspect before paying. Step up to M1 only if the twenty-organisation case is firm and the FP8 headroom is genuinely wanted.

What would change the answer

Measured concurrency above roughly twenty in flight, or sessions routinely running at 32k context rather than 16k. Either doubles the KV cache and 48 GB stops being enough. **A large dense model entering the roadmap** — everything here assumes a sparse mixture-of-experts agent in the 30B class, and a dense 70B is a 96 GB conversation. **A hosted open-weight API being acceptable**: GLM-5.2 serves this workload for AED 1,270 a month against AED 1,369 owned, so if documents may leave at all, nothing here beats it below about sixteen clients.

The market you are buying into

One global condition, and three things about buying it in the UAE.

Any hardware bought in 2026 is bought during a memory shortage, and that fact governs every price in this report. It is not a normal market and the usual instincts — buy new, buy the current generation, wait for prices to fall — are all wrong in specific ways worth naming before a bill of materials is read.

The global condition

The AI datacentre buildout is absorbing ~20% of global DRAM capacity absorbed by AI, and the allocation is not neutral between product lines: DDR5 wafers go to server memory, GDDR7 goes to accelerators, and consumer parts get what is left. NVIDIA reportedly considering 30-40% output cuts on mid-range parts to prioritise higher-margin AI silicon. Published forecasts do not expect meaningful relief before 2028.

The visible result is that prices rose without performance changing. An RTX PRO 6000 listed at USD 8,565 when it launched in March 2025 and USD 16,000 by August 2026 — an 87% increase on the same silicon. RTX 5090 street pricing runs USD 4,300 to 5,000 against a USD 1,999 launch price. **Nothing about those cards got faster. The memory they are made of got scarce.**

The consequence for this decision

When the price of owning capacity rises for reasons unrelated to performance, two things follow. Buying at the top of a squeeze carries mark-to-market risk that buying in a normal market does not — if supply normalises, the card does not get slower but its replacement cost falls. And the second-hand market, where prices are set by people selling cards rather than by fabs allocating wafers, becomes the one segment still priced on merit. Both point the same way: **buy used, buy small, or rent.**

Three things about buying it here

Local retail carries a real premium, and a wide spread. An RTX 5090 Founders Edition sits above AED 12,000 in the UAE against a USD 1,999 (about AED 7,340) launch price. More usefully for planning, the spread *between UAE sellers* on one part number is enormous: the RTX PRO 6000 ranges from AED 48,825 to AED 77,032 depending on the vendor, and Promise Gulf’s own product page and price list disagree by more than a factor of two on the PRO 5000. **Every four- and five-figure line in this report should be re-quoted in writing before it is committed to**, and the spread itself is a negotiating position.

The second-hand market here is deep enough to build on. Dubizzle lists roughly 290 used graphics cards nationally with a further 91 in Abu Dhabi — a real market rather than a handful of listings. RTX 3090s appear around AED 2,000 to 2,300, which is a quarter of what a used 4090 costs locally and about a sixth of a new 5090. That single price relationship is why the low tier of this report is competitive with the high tier on capability, and it is a specifically local advantage.

Buying locally removes two cost lines entirely. Importing hardware means 5% customs duty on the CIF value, 5% VAT on top of that, freight and a Mirsal 2 clearance. Buying from a UAE seller means the duty is already inside the shelf price, there is no freight to arrange and no clearance to wait on, and the only adjustment is the 5% VAT shown on the tax invoice, which a VAT-registered business recovers. Every build in this report is therefore costed at a listed AED price plus local delivery, with VAT stripped to give the effective capital cost.

	Import it	Buy it here
Customs duty	5% on CIF value	already inside the shelf price
VAT	5% on CIF plus duty, paid at clearance	5% on the invoice, recovered if registered
Freight	international, plus Mirsal 2 clearance	local delivery, about AED 150

	Import it	Buy it here
Lead time	weeks, and a customs process	next-day from most Dubai sellers
Warranty	an international RMA, at your cost	local warranty on new stock; none at all on a private sale

The comparison matters because a report priced from US retail would add roughly 10% in duty and VAT plus freight to every build, and would be describing a purchase nobody here needs to make.

What this means for the tiers

Strategy	In this market	Verdict
Buy new consumer silicon	Paying shortage prices for VRAM. A new 16 GB card and two used 24 GB cards cost the same here.	Only for warranty and simplicity, not for capacity
Buy used consumer silicon	Priced by sellers, not by allocation. Deep local supply on Ampere.	The best value available, if a private sale is acceptable
Buy workstation silicon	Rose on supply, not on performance, and carries the widest local price spread.	Defensible only with a written quote at the low end of the range
Rent internationally	Providers competing on utilisation of assets bought before the squeeze.	Cheapest per hour, forfeits residency
Rent in region	No consumer-class option exists; datacentre hardware only.	About four times the international rate, keeps data in the country

From client organisations to gigabytes

An auditable chain: traffic, then concurrency, then memory. Change any input and the rest re-derives.

“Ten to twenty client organisations” is a commercial target, not an engineering one. It has to be turned into a number of simultaneously in-flight requests before it can be turned into a card, and the gap between those two numbers is large enough to change which tier gets bought.

Step 1 — the workload

Four slots run on this machine, and one of them is free. The agent is a tool-calling language model. Vision is a separate model that reads scanned invoices and images. Speech in and speech out are two more. **Graph generation is not a model at all** — it is matplotlib writing a PNG, or reportlab writing a chart into a PDF, and it belongs in `app/tools.py` next to `write_excel` as an ordinary Python function. It costs no VRAM, no weights and no load time. Building it as a diffusion model would be the single most expensive mistake available in this project.

Slot	Model class	Serving quant	VRAM	Note
Agent	Qwen3-Coder-30B-A3B	INT4 / AWQ	18 GB	MoE, ~3B active per token — large-model reasoning at small-model decode speed
Vision	Qwen3-VL-8B	FP8	9 GB	Native-resolution encoder; the practical single-GPU choice for document OCR
Speech in	Whisper Large V3 Turbo	FP16 / CTranslate2	3 GB	809M params, ~216× real time; a naive load costs 6 GB, faster-whisper costs 3
Speech out	Kokoro-82M	FP16	1 GB	Runs on CPU. Do not give it a GPU slot until measurement says otherwise
Graphs	matplotlib / reportlab	—	0 GB	A CPU tool, not a model. The cheapest requirement in the whole brief

Resident weights total 31 GB. That is the floor, and it is the smaller half of the problem.

Step 2 — traffic, and why organisations are not concurrency

Each organisation is assumed to have 4 active users, each running 12 agent sessions a day of 4 turns. Each turn sends about 6,000 prompt tokens and receives 400 — the prompt figure is deliberately high because `app/agent.py` re-sends the entire conversation plus every tool result on every step, and **TOOL_RESULT_BUDGET** alone is 12,000 characters. A fifth of the day’s turns are assumed to land in the busiest hour, which is a normal office shape and a pessimistic one.

A turn occupies a slot for about 18 seconds: roughly 2.4s of prefill at a modelled 2,500 tok/s share under batching, then 16s of decode at the 25 tok/s per user that reads as responsive. Offered load is arrivals times service time; the number of slots needed to keep the wait probability under 10% comes from Erlang C.

Orgs	Users	Turns/day	Peak hour	Offered load	Slots needed	P(wait)	Tokens/month
5	20	960	192	0.98 E	3	9%	135 M
10	40	1,920	384	1.96 E	5	6%	270 M
15	60	2,880	576	2.94 E	6	9%	406 M
20	80	3,840	768	3.93 E	8	5%	541 M
30	120	5,760	1,152	5.89 E	10	9%	811 M
40	160	7,680	1,536	7.85 E	13	7%	1081 M

The design row is fifteen organisations. Note the shape: quadrupling the client count from five to twenty roughly triples the token bill but only raises required concurrency from three slots to eight. Statistical multiplexing is why a small machine serves a lot of people.

Twenty organisations is eight in-flight requests, not twenty

This is the single most consequential number in the report. Sizing to twenty concurrent sessions would demand roughly 70 GB and push the decision into the high tier. Sizing to the eight that queueing theory actually calls for lands at 50 GB and makes a pair of used cards sufficient. The difference between those two readings is about ten thousand dollars.

Step 3 — concurrency into memory

The KV cache is the state a model keeps for a conversation in progress, and it is the term that scales with users rather than with model size:

KV bytes per token = $2 \times \text{layers} \times \text{kv_heads} \times \text{head_dim} \times \text{dtype_bytes}$

The 2 is one tensor for keys and one for values. **kv_heads**, not **heads**, is what matters — grouped-query attention shares key/value projections across attention heads, and it is the single reason a 30B model can hold twenty sessions on one card. Multiply by context length and by concurrent sessions to get the total.

The formula is checked against this project's own measurements rather than taken on faith.

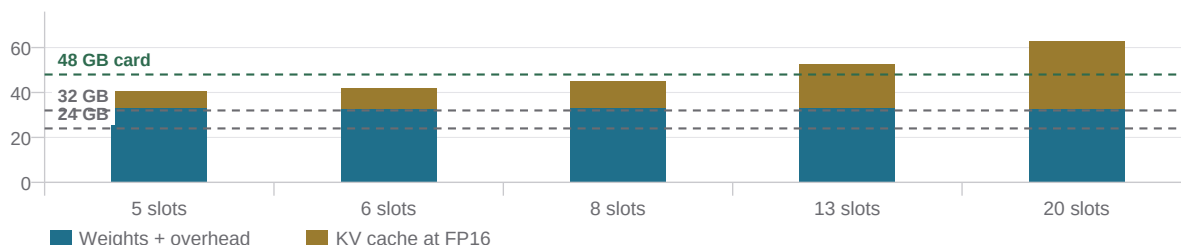
bench_results.json, measured on 2 September on a temporary RTX 3060 test machine — a box used to build the software, not part of any option in this report and not counted as an asset anywhere in it:

Model	Context	Predicted	Measured	Ratio	
qwen3:8b	8,192	1,152 MiB	986 MB	0.86	within measurement error of the formula
qwen3:8b	16,384	2,304 MiB	2,157 MB	0.94	within measurement error of the formula
qwen3:14b	8,192	1,280 MiB	982 MB	0.77	GQA sharing plus allocator rounding
qwen3:14b	16,384	2,560 MiB	1,236 MB	0.48	card at its ceiling — throughput collapsed to 8.25 tok/s on this run, so the reading is offloading, not cache

The two 8B rows agree with the model to within 15%, which is as close as an allocator-level measurement gets. The final row is excluded from the fit: at 10.9 GB peak on a 12 GB card, that run was paging, and its collapse from 31.7 to 8.25 tok/s is the evidence.

The budget

A 30B-A3B-class agent is modelled at 48 layers, 4 KV heads and 128 head dimension — stated as an assumption, not measured; the 8B above validates the formula, not this configuration. At 16k context that is 96 KiB per token at FP16 and half that at FP8, and vLLM is given 90% of the card rather than all of it.



Stacked VRAM requirement against concurrent slots, with card sizes marked. Note where the crossings fall: 24 GB is out at any concurrency once a vision model is resident, 32 GB runs out around six slots, and 48 GB carries the twenty-organisation case with room to spare.

Concurrent slots	Serves	Weights	KV @ FP16	KV @ FP8	Card needed, FP16	Card needed, FP8
5	10 orgs	31 GB	7.5 GB	3.8 GB	45 GB	41 GB
6	15 orgs	31 GB	9.0 GB	4.5 GB	47 GB	42 GB
8	20 orgs	31 GB	12.0 GB	6.0 GB	50 GB	43 GB
13	40 orgs	31 GB	19.5 GB	9.8 GB	58 GB	48 GB
20	a bad day	31 GB	30.0 GB	15.0 GB	70 GB	53 GB

FP8 KV cache is available on Ada and Blackwell, not on Ampere. That single capability is worth roughly 5 GB at the design point and is the strongest technical argument for buying 4090s rather than 3090s.

The honest caveat

Every measured number in this project is a single request at a time on Ollama, which does not batch. The formula above is validated against those measurements; the concurrency figures are **modelled, not measured**. Appendix A sets out what to instrument, and section 8 puts a two-week rental in front of any purchase precisely so this paragraph can be deleted from the next revision.

The serving layer: Ollama today, vLLM next

The highest-leverage change on the list, and the only one that costs nothing.

Ollama was the right choice for this project and remains the right choice for a one-person box: same commands on Windows and Linux, a model pulled in one line, and a cold start of 2.1s against vLLM's 4.8s. **app/llm.py** talks to it in a single standard-library POST, and the README is right that a client library would only fall out of step with a fast-moving server.

What it does not do is serve several people at once. Requests are handled one after another. On a single stream the difference is modest — 197 against 178 tok/s on the same card and quantisation — but that comparison measures the wrong thing. Under load, a batching server reaches roughly 850 tok/s aggregate at batch 32, which is about 30 users each still seeing 25 tok/s. The serial server has no equivalent number, because the second user waits for the first.

Why batching is nearly free, and why image generation is not

Generating a token reads the model's weights out of memory. Several conversations in the same batch **share that read**, so the marginal cost of the second user is arithmetic, not bandwidth — and bandwidth is the bottleneck. That is the whole trick. It is also why **app/jobs.py** exists and is correct: diffusion does real work per denoising step and does not share, so it queues rather than batches. One batches, one queues. Keeping those two behaviours in separate lanes is a design decision this project already made and should keep.

Three ceilings, not one

Swapping the inference server alone will not deliver concurrency, because two other limits sit in front of it. All three have to lift together, and none of them costs money:

#	Ceiling	Where	Change
1	Serial inference	app/llm.py → Ollama /api/chat	Point it at an OpenAI-compatible /v1/chat/completions endpoint. The file is one POST; step-00 estimates about an hour.
2	One job worker	JOB_WORKERS=1 in .env	Correct today, and documented as correct: one GPU serialises the work anyway. It becomes wrong the moment a second card or a batching server arrives.
3	One web process	uvicorn.run in app/main.py, no --workers	Single-process uvicorn will saturate on request handling long before the GPU does at eight concurrent agent turns.

Buying a larger card without lifting all three buys latency on one request and nothing on ten. This is the cheapest work in the report and it gates every hardware decision after it.

What it costs: the operating system

vLLM has no native Windows support. The machine that holds the production card should run Ubuntu. WSL2 is fine for development but adds a virtualised memory boundary that is unhelpful precisely when the card is full. This is not a surprise and not a rewrite: **scripts/service/syslab-server.service** already exists as the systemd equivalent of the Windows scheduled tasks, every path goes through **pathlib**, and every setting lives in **.env**. The README calls those two habits the reason the Linux move is a copy of the folder rather than a rewrite. They are about to be tested, and the plan should assume a Windows development box and a Linux serving box from here on.

Prefix caching is worth more here than on most workloads

vLLM's automatic prefix caching keeps the KV blocks of a shared prompt prefix and reuses them across requests. Most chat products get a modest win from it. This one should get an unusually large one, because **app/agent.py** re-sends the entire conversation plus every tool result on every step of the loop: turn four of a session is turn three's prompt with a tool result appended. The system prompt and twelve tool descriptions are identical on every request from every tenant. On a workload whose prompt-to-output ratio is roughly fifteen to one, cutting repeated prefill is worth more than any plausible card upgrade — and it costs a configuration flag.

The alternatives, assessed rather than dismissed

Runtime	Batching	Verdict for this project
vLLM	Continuous, PagedAttention	The recommendation. Best-supported OpenAI-compatible server, widest quantisation coverage, automatic prefix caching. Linux only.
SGLang	Continuous, RadixAttention	A real contender, and arguably better suited: RadixAttention shares prefix cache across a tree of requests, which is the exact shape of an agent loop. Worth benchmarking during the rental fortnight rather than assumed away.
TensorRT-LLM	Continuous	Fastest on NVIDIA silicon and the worst ergonomics here. An engine must be compiled per model, per quantisation, per GPU. Wrong trade for a project that will change models.
llama.cpp	Limited	The only runtime that does MoE expert offload to system RAM well — the technique behind the 120B-on-12 GB result. Keep it in the toolbox for that specific case; do not serve clients from it.
Ollama	None	Keep it on the development box. It is the reason the project got built quickly and it remains the fastest way to try a new model. It is not a serving tier.

Quantisation, and the metric that actually matters

The temptation is to pick the quantisation that fits and move on. For this project the constraint is different: the model's job is to emit well-formed tool calls, and tool-calling accuracy degrades under aggressive quantisation earlier and less gracefully than prose quality does. A model that writes a slightly worse sentence is fine. A model that omits a required argument sends the loop round again, and **MAX_TOOL_STEPS** is 10.

Scheme	Hardware	Memory	Note
FP8 weights + FP8 KV	Ada, Blackwell	~50% of FP16	The right default on 4090 / 5090 / PRO class. Near-lossless in practice and it halves the KV cache, which is where the pressure is.
AWQ / GPTQ INT4	Ampere and later	~25% of FP16	What Ampere gets, because it has no FP8 path. Calibrated 4-bit holds tool calling well; verify against the project's own <code>check_agent.py</code> rather than a published score.
Q4_K_M (GGUF)	any, llama.cpp/Ollama	~28% of FP16	What the project runs today. Fine for a single stream, not a vLLM serving format.
FP16	any	100%	Only worth it if a regression appears that quantisation explains. Costs roughly twice the card for no measured benefit at this scale.

Test quantisation with the gates that already exist

This project has **check_agent.py**, **check_tools.py**, **check_search.py**, **check_database.py** and a 279-test pytest suite, all of which assert on tool calls returning **ok** rather than merely being attempted. That is a better quantisation acceptance test than any leaderboard, because it measures the thing the product sells. Run the gates at each candidate quantisation and pick the smallest one that stays green.

Model selection

Four slots, four shortlists, and one licence trap worth naming twice.

The recommended agent is Qwen3-Coder-30B-A3B-Instruct, on a 92% Berkeley Function Calling score and an Apache 2.0 licence. The reasoning behind it matters more than the specific name: **a sparse mixture-of-experts model in the 30B class is the right shape for this workload**. It occupies the memory of a large model but reads only about three billion parameters per token, so it reasons like something big and decodes like something small. On a bandwidth-bound machine that is the entire game.

The field moved during 2026 — GLM, Qwen, MiniMax and Nemotron all shipped iterations — but the shortlist below is deliberately conservative. Model names churn faster than a procurement cycle, and this project has something better than a leaderboard to choose with: its own gates.

Agent — the tool-calling model

Model	Shape	VRAM INT4	VRAM FP8	Licence	Tool use
Qwen3-Coder-30B-A3B-Instruct	30B total / 3B active, MoE	18 GB	32 GB	Apache 2.0	92% BFCL
GLM-4.7-Flash	31B dense/hybrid	17 GB	34 GB	Apache 2.0	strong, reasoning + tool use
Qwen3-32B	32B dense	16 GB	32 GB	Apache 2.0	good
Nemotron Nano 9B	9B dense	6 GB	10 GB	NVIDIA Open Model	95% on a local tool-calling suite
qwen3:8b	8.2B dense	5 GB	9 GB	Apache 2.0	adequate with the loop's guardrails

Qwen3-Coder-30B-A3B-Instruct. The recommended agent. MoE means it reads about 3B of weights per token, so it decodes at the speed of a small model while reasoning like a large one - which is the whole game on a bandwidth-bound machine.

GLM-4.7-Flash. 68 GB at FP16, 34 GB at INT8, 17 GB at INT4. Listed among the strongest models that fit inside 32 GB.

Qwen3-32B. Dense: every parameter is read per token, so it decodes roughly ten times slower than the 30B-A3B on the same card. Included as the counter-example that makes the MoE case.

Vision — reading images and scanned documents

Model	VRAM	Licence	Note
Qwen3-VL-8B	9 GB	Apache 2.0	The practical single-GPU choice. Native-resolution vision encoder, strong document OCR and table extraction.
Qwen3-VL-30B-A3B	18 GB	Apache 2.0	MoE vision model. Worth the slot only if invoice-field extraction becomes the product.
Qwen3-VL-4B	3 GB	Apache 2.0	Fits anywhere. The right first VLM to ship.
GLM-4.5V	24 GB	commercial-friendly	Thinking modes for deep document analysis; heavier than the workload warrants.

Start with the 4B, budget for the 8B. A vision model earns a permanent slot only once field extraction from invoice PDFs is a product feature rather than an experiment — and HANDOVER.md already names structured field extraction as the highest-value unbuilt thing.

Speech in and speech out

Slot	Model	VRAM	Licence	Note
Speech in	Whisper Large V3 Turbo (809M)	3 GB	MIT	Decoder pruned from 32 layers to 4; about 216x real time, WER 7.75% against 7.4% for full large-v3. Runs in 2-3 GB under faster-whisper rather than the 6 GB a naive load costs.
Speech in	NVIDIA Parakeet TDT	2 GB	NVIDIA open	2,728x RTFx - the fastest self-hostable option for batch throughput. English-first.
Speech in	Voxtral Transcribe 2	5 GB	open	5.9% average WER against Whisper's 7.4% - the accuracy leader if transcription quality becomes contractual.
Speech out	Kokoro-82M	1 GB	Apache 2.0	Runs on CPU. At 82M parameters this is the cheapest slot in the stack and should not be given a GPU until measurement says otherwise.
Speech out	Chatterbox	5 GB	open	Better prosody, five times the memory. Only if voice quality is a selling point.

Voice is the cheapest capability in the brief after graph generation. Whisper Turbo and Kokoro together fit in 4 GB and can largely run beside the agent rather than competing with it.

Graph generation is not a model

Worth stating once more in its own section, because it is the requirement most likely to be mis-specified into a hardware line item. “Generate a graph of monthly removals by site” is a `run_sql` call followed by a `matplotlib` call. The model chooses the query and the chart type; Python draws the picture. It belongs in `app/tools.py` beside `write_excel` and `write_pdf`, as a plain function with no knowledge that a model exists — which is exactly the contract the README sets for every tool in that file. Zero VRAM, testable without a GPU, and deterministic.

Licences: read the file, not the family name

Read the LICENSE file, not the family name

FLUX.2 klein-4B shipped Apache 2.0 and klein-9B shipped non-commercial, on the same day, under the same family name. Only one of them is sellable. The rule that follows is cheap to apply and expensive to skip: **before shipping anything to a paying client, open the LICENSE file in the exact repository and exact revision being deployed.** Community licences with user-count caps, research-only terms and revenue thresholds all exist inside families whose other members are permissive.

For this project the practical consequence is narrow. Apache 2.0 and MIT cover the entire recommended stack — agent, vision, speech in, speech out and charting. There is no need to accept a restrictive licence to build what has been specified, and no reason to.

What to actually run, by tier

If the machine has	Agent	Vision	Voice	Comment
12–24 GB	qwen3:8b or Nemotron Nano 9B, INT4	Qwen3-VL-4B	Whisper Turbo, Kokoro on CPU	Today's stack. One or two users, no batching headroom.
32 GB	Qwen3-Coder-30B-A3B, INT4	Qwen3-VL-4B	Whisper Turbo, Kokoro on CPU	The agent fits; the KV cache is what runs out. FP8 cache is mandatory here.
48 GB	Qwen3-Coder-30B-A3B, INT4 or FP8	Qwen3-VL-8B	Whisper Turbo, Kokoro	The design point. Everything resident, eight slots at 16k, room to grow.

If the machine has	Agent	Vision	Voice	Comment
96 GB	Qwen3-Coder-30B-A3B at FP8, or a 70B-class dense model	Qwen3-VL-30B-A 3B	Voxtral, Chatterbox	Buys model headroom rather than user headroom. Only worth it if the model roadmap changes.

Nine options, three tiers

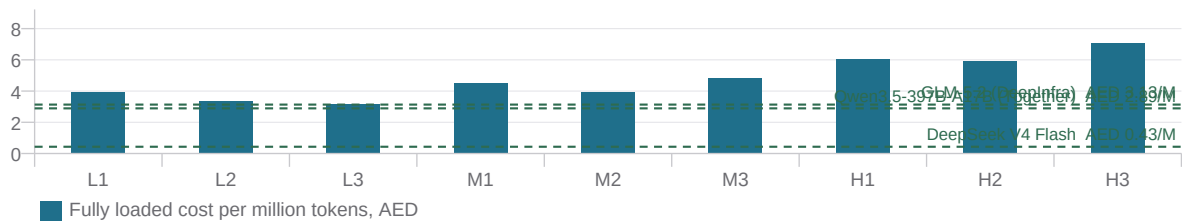
Every price is a UAE seller's listed AED price, VAT-inclusive, plus local delivery. No duty, no freight.

The tiers are budget bands, not a ladder: moving up one does not strictly dominate the band below. New and second-hand options appear in every tier deliberately, because whether a private sale is acceptable is a separate decision from how much to spend, and the two should be made independently rather than bundled.

At a glance

	Option	Cond.	VRAM	Delivered	All-in / mo	AED/M tok	Design point
L1	Rent, own nothing	rent	—	—	AED 1,598	3.94	Serves 20 orgs
L2	Two used RTX 3090 - 48 GB	used	48 GB	AED 15,569	AED 1,369	3.38	Serves 15, tight at 20
L3	One new RTX 5080 - 16 GB	new	16 GB	AED 15,895	AED 1,294	3.19	Under-sized
M1	Two used RTX 4090 - 48 GB	used	48 GB	AED 29,769	AED 1,835	4.53	Serves 20 orgs
M2	One new RTX 5090 - 32 GB	new	32 GB	AED 25,769	AED 1,609	3.97	Under-sized
M3	Four used RTX 3090 - 96 GB	used	96 GB	AED 31,089	AED 1,971	4.86	Serves 20 orgs, headroom
H1	Two new RTX 5090 - 64 GB	new	64 GB	AED 51,489	AED 2,450	6.04	Serves 20 orgs
H2	Two RTX PRO 5000 Blackwell - 96 GB ECC	new	96 GB	AED 54,919	AED 2,415	5.95	Serves 20 orgs, headroom
H3	One RTX PRO 6000 Blackwell - 96 GB ECC	new	96 GB	AED 71,919	AED 2,879	7.10	Serves 20 orgs, headroom

All-in monthly is capital over 36 months plus electricity, cooling and four hours of operator time, at 12 hours of GPU duty a day. The rent row uses the international line; its in-region equivalent is on its own page and in section 7.



Owned cost per million tokens against API list prices converted to AED, at the 406 M tokens a month the design point generates. Frontier APIs sit far above every option here and are off the top of the scale; the open-weight hosts sit among them.

How to read the nine pages that follow

Each option gets a bill of materials priced line by line against a named UAE seller, a capability check against the design point from section 3, and — for the eight that involve buying something — a buy-versus-rent table at eight, twelve and twenty-four hours of daily GPU duty, computed twice: once against renting abroad and once against renting in the UAE. A break-even beyond 36 months is shown in red, which is the polite way of saying the machine never repays itself.

LOW TIER · OPTION L1 · NO CAPITAL

Rent, own nothing

CAPITAL AED 0	INTERNATIONAL AED 1,598/mo	IN REGION AED 4,067/mo	PER M TOKENS 3.94	DESIGN POINT Serves 20 orgs
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Port app/llm.py to the OpenAI-compatible shape and serve from a rented instance. There is no machine to build, no card to source in a shortage, no warranty to argue about and no capital at risk. The whole question is where the card sits. Internationally, a 48 GB card runs about \$0.64 an hour and the economics are excellent. In the UAE there is no consumer-class rental at all: the cheapest in-region option is a datacentre H100 at \$2.48, roughly four times the price for hardware far larger than this workload needs. That gap is the price of keeping documents in the country, and it is quantified on this page rather than assumed away. Development also has to live somewhere under this option, since no machine is bought.

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	48 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	n/a — an operating cost	✓

Verdict

Zero capital, and the only option whose cost falls if the client count turns out smaller than planned. Whether it is admissible at all depends on one answer nobody has yet.

What it cannot do. On the international line, client documents leave the UAE. If residency is contractual that line is inadmissible at any price, and the in-region line is the only rental that counts.

LOW TIER · OPTION L2 · SECOND-HAND

Two used RTX 3090 - 48 GB

DELIVERED AED 15,569	VRAM 48 GB	ALL-IN MONTHLY AED 1,369	PER M TOKENS 3.38	DESIGN POINT Serves 15, tight at 20
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Two matched 24 GB cards, so vLLM can run tensor parallel and treat the 48 GB as one pool. This build exists because of a local price relationship: RTX 3090s list on Dubizzle around AED 2,000 to 2,300, roughly a quarter of a used 4090 here and a sixth of a new 5090, while carrying over half a 5090's bandwidth at 936 GB/s. Inspect before paying - power-on hours under about 20,000, a sustained load test in front of the seller, and a meeting place where the card can be plugged in. The 12% failure reserve is there because none of that is a warranty.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 3090 24 GB, used ×2	Dubizzle	AED 2,400	AED 4,800
AMD Ryzen 9 9950X (16C, AM5)	GCC Gamers	AED 2,200	AED 2,200
X870E ATX, dual x16 slots (x8/x8)	Microless	AED 1,900	AED 1,900
64 GB DDR5 (2x32)	CostToCost	AED 2,400	AED 2,400
2 TB NVMe Gen4	Microless	AED 1,070	AED 1,070
Full-tower case + fans	—	AED 500	AED 500
360 mm AIO	—	AED 550	AED 550
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1200 W 80+ Gold/Platinum	Alershad	AED 899	AED 899
Retail subtotal			AED 15,419
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 15,569
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 576 failure reserve)			AED 15,404

Capability against the design point — 47 GB at 15 organisations, 50 GB at 20

Capability	This option	
Pooled memory vLLM can use	48 GB	✗
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	no — Ampere; cache stays full width	✗
Warranty and tax invoice	no — private sale	✗

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 5000 48 GB, \$0.64/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 572	AED 2,218	AED 906	38 mo	8 mo	388 W
12 h/day	AED 858	AED 3,327	AED 941	23 mo	5 mo	470 W
24 h/day	AED 1,717	AED 6,653	AED 1,046	11 mo	2 mo	715 W

Verdict

The cheapest hardware that reaches the design point, by a wide margin. The UAE second-hand market is what makes it possible.

What it cannot do. No FP8. Ampere has no 8-bit path, so the KV cache stays full width and the 48 GB that comfortably serves fifteen organisations is tight at twenty. There is also no warranty and no tax invoice on a private sale.

LOW TIER · OPTION L3 · NEW, WARRANTED

One new RTX 5080 - 16 GB

DELIVERED AED 15,895	VRAM 16 GB	ALL-IN MONTHLY AED 1,294	PER M TOKENS 3.19	DESIGN POINT Under-sized
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Included because 'buy something new, from a shop, with a receipt' is a reasonable procurement instinct and deserves a real answer. At the same money as two used 3090s this build gets a third of the memory. That is not a criticism of the 5080, which is a fine card; it is what the shortage has done to the price of new VRAM. Its honest role is a development and staging machine beside a rented production endpoint, or a first box for a single operator - not a server behind fifteen clients.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 5080 16 GB, new	Amazon.ae	AED 5,575	AED 5,575
AMD Ryzen 9 9950X (16C, AM5)	GCC Gamers	AED 2,200	AED 2,200
X870E ATX, dual x16 slots (x8/x8)	Microless	AED 1,900	AED 1,900
64 GB DDR5 (2x32)	CostToCost	AED 2,400	AED 2,400
2 TB NVMe Gen4	Microless	AED 1,070	AED 1,070
Full-tower case + fans	—	AED 500	AED 500
360 mm AIO	—	AED 550	AED 550
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
850 W 80+ Gold	—	AED 450	AED 450
Retail subtotal			AED 15,745
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 15,895
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 167 failure reserve)			AED 15,305

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	16 GB	✗
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	yes, local warranty	✓

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX 5090 32 GB, \$0.33/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 295	AED 2,218	AED 850	83 mo	7 mo	258 W
12 h/day	AED 443	AED 3,327	AED 868	49 mo	5 mo	300 W
24 h/day	AED 885	AED 6,653	AED 922	22 mo	2 mo	426 W

Verdict

The cheapest rig you can buy new, with a warranty and a tax invoice - and it does not meet the specification. That is the finding, not a flaw in the build.

What it cannot do. 16 GB holds the agent or the vision model, not both, and leaves almost nothing for KV cache. It cannot serve ten organisations, let alone twenty.

MID TIER · OPTION M1 · SECOND-HAND

Two used RTX 4090 - 48 GB

DELIVERED AED 29,769	VRAM 48 GB	ALL-IN MONTHLY AED 1,835	PER M TOKENS 4.53	DESIGN POINT Serves 20 orgs
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Ada generation means native FP8, which halves the KV cache and turns 48 GB from adequate at fifteen organisations into comfortable at twenty. It is a genuine capability difference and the question is only whether it is worth the gap. Locally that gap is large: used 4090s run AED 8,300 to 12,000 against AED 2,000 to 2,300 for a 3090, so the pair costs about four times as much for the same memory and one quantisation feature. Buy this if the twenty-organisation case is firm and you want headroom; buy L2 if it is still a projection.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 4090 24 GB, used or renewed ×2	Microless	AED 9,500	AED 19,000
AMD Ryzen 9 9950X (16C, AM5)	GCC Gamers	AED 2,200	AED 2,200
X870E ATX, dual x16 slots (x8/x8)	Microless	AED 1,900	AED 1,900
64 GB DDR5 (2x32)	CostToCost	AED 2,400	AED 2,400
2 TB NVMe Gen4	Microless	AED 1,070	AED 1,070
Full-tower case + fans	—	AED 500	AED 500
360 mm AIO	—	AED 550	AED 550
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1200 W 80+ Gold/Platinum	Alershad	AED 899	AED 899
Retail subtotal			AED 29,619
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 29,769
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 2,280 failure reserve)			AED 30,631

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	48 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	no — private sale	✗

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 5000 48 GB, \$0.64/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 572	AED 2,218	AED 939	82 mo	15 mo	465 W
12 h/day	AED 858	AED 3,327	AED 984	50 mo	10 mo	570 W
24 h/day	AED 1,717	AED 6,653	AED 1,119	23 mo	5 mo	885 W

Verdict

The same 48 GB as L2 with the FP8 path that removes its one real weakness - for roughly twice the money.

What it cannot do. Still a private-sale purchase with no warranty, at a price where that matters considerably more than it does on a AED 2,400 card.

MID TIER · OPTION M2 · NEW, WARRANTED

One new RTX 5090 - 32 GB

DELIVERED AED 25,769	VRAM 32 GB	ALL-IN MONTHLY AED 1,609	PER M TOKENS 3.97	DESIGN POINT Under-sized
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One card, one warranty, one supplier, and 1,792 GB/s - the fastest single-request experience in this report and the simplest machine to operate. It is also the clearest illustration of what the shortage costs: AED 13,839 to 22,000 for 32 GB, against AED 4,800 for 48 GB in used Ampere. This workload spends most of a turn on 6,000 tokens of prefill and tool round-trips, so it feels the extra bandwidth far less than a single-user chat product would. Buy it if warranty and simplicity are the requirement; do not buy it for capacity.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 5090 32 GB, new	GCC Gamers	AED 15,000	AED 15,000
AMD Ryzen 9 9950X (16C, AM5)	GCC Gamers	AED 2,200	AED 2,200
X870E ATX, dual x16 slots (x8/x8)	Microless	AED 1,900	AED 1,900
64 GB DDR5 (2x32)	CostToCost	AED 2,400	AED 2,400
2 TB NVMe Gen4	Microless	AED 1,070	AED 1,070
Full-tower case + fans	—	AED 500	AED 500
360 mm AIO	—	AED 550	AED 550
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1200 W 80+ Gold/Platinum	Alershad	AED 899	AED 899
Retail subtotal			AED 25,619
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 25,769
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 450 failure reserve)			AED 24,992

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	32 GB	✗
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	yes, local warranty	✓

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX 5090 32 GB, \$0.33/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 295	AED 2,218	AED 886	167 mo	12 mo	340 W
12 h/day	AED 443	AED 3,327	AED 914	93 mo	8 mo	408 W
24 h/day	AED 885	AED 6,653	AED 1,001	40 mo	4 mo	609 W

Verdict

The best new card most UAE sellers will actually hand you today, in the wrong size for this job.

What it cannot do. 32 GB will not hold the agent, a vision model and eight sessions of KV cache at 16k context without quantising the cache to FP8 and dropping to the 4 GB vision model.

MID TIER · OPTION M3 · SECOND-HAND

Four used RTX 3090 - 96 GB

DELIVERED AED 31,089	VRAM 96 GB	ALL-IN MONTHLY AED 1,971	PER M TOKENS 4.86	DESIGN POINT Serves 20 orgs, headroom
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Four cards at AED 2,400 is AED 9,600 for 96 GB - the cheapest large-memory CUDA machine that can be assembled anywhere, and the UAE used market makes it cheaper here than most places. The catch is everything around them: an sTR5 platform to give four cards their lanes costs more than the cards do, and 128 GB of registered memory is now one of the largest lines in the build. Take this option if the roadmap genuinely lands on a large dense model or a resident 30B vision model. For the workload as specified it is buying capacity that will sit unused.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 3090 24 GB, used x4	Dubizzle	AED 2,400	AED 9,600
AMD Ryzen Threadripper 7960X (24C, sTR5, 88 lanes)	GCC Gamers	AED 6,150	AED 6,150
Gigabyte TRX50 AERO D, sTR5, 4x x16	—	AED 2,600	AED 2,600
128 GB DDR5 RDIMM (TRX50, quad channel)	—	AED 6,000	AED 6,000
4 TB NVMe Gen4	Microless	AED 2,020	AED 2,020
Full-tower case + fans	—	AED 500	AED 500
sTR5 air cooler	—	AED 500	AED 500
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1600 W 80+ Titanium	Alershad	AED 1,819	AED 1,819
PCIe 4.0 risers + GPU mounts	—	AED 650	AED 650
Retail subtotal			AED 30,939
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 31,089
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 1,152 failure reserve)			AED 30,761

Capability against the design point — 47 GB at 15 organisations, 50 GB at 20

Capability	This option	
Pooled memory vLLM can use	96 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	no — Ampere; cache stays full width	✗
Warranty and tax invoice	no — private sale	✗

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 6000 96 GB, \$2.20/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 1,967	AED 2,218	AED 1,047	19 mo	16 mo	717 W
12 h/day	AED 2,951	AED 3,327	AED 1,117	12 mo	10 mo	880 W
24 h/day	AED 5,902	AED 6,653	AED 1,327	6 mo	5 mo	1370 W

Verdict
96 GB for less than the price of one new 5090's worth of cards. The platform, not the silicon, is what you are paying for.

What it cannot do. No FP8, four unwarranted cards, and 1.4 kW under load. NVLink pairs only two of the four, so tensor parallel across all of them crosses PCIe.

HIGH TIER · OPTION H1 · NEW, WARRANTED

Two new RTX 5090 - 64 GB

DELIVERED AED 51,489	VRAM 64 GB	ALL-IN MONTHLY AED 2,450	PER M TOKENS 6.04	DESIGN POINT Serves 20 orgs
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Two cards means the agent and the vision lane never contend: one holds the tool-calling model and its cache, the other holds the VLM and speech, and a slow document parse cannot make a chat turn wait. That is the argument app/jobs.py already makes in software, expressed in hardware. It is the first build here that is both fully warranted and comfortably over the design point, which is worth something to a business signing client agreements. Whether it is worth AED 30,000 of graphics card is the question section 7 answers.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX 5090 32 GB, new x2	GCC Gamers	AED 15,000	AED 30,000
AMD Ryzen Threadripper 7960X (24C, sTR5, 88 lanes)	GCC Gamers	AED 6,150	AED 6,150
Gigabyte TRX50 AERO D, sTR5, 4x x16	—	AED 2,600	AED 2,600
128 GB DDR5 RDIMM (TRX50, quad channel)	—	AED 6,000	AED 6,000
4 TB NVMe Gen4	Microless	AED 2,020	AED 2,020
Full-tower case + fans	—	AED 500	AED 500
sTR5 air cooler	—	AED 500	AED 500
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1600 W 80+ Titanium	Alershad	AED 1,819	AED 1,819
PCIe 4.0 risers + GPU mounts	—	AED 650	AED 650
Retail subtotal			AED 51,339
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 51,489
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 900 failure reserve)			AED 49,937

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	64 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	yes, local warranty	✓

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 6000 96 GB, \$2.20/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 1,967	AED 2,218	AED 1,006	29 mo	26 mo	621 W
12 h/day	AED 2,951	AED 3,327	AED 1,063	19 mo	17 mo	755 W
24 h/day	AED 5,902	AED 6,653	AED 1,236	9 mo	8 mo	1158 W

Verdict

Buys lane isolation and the fastest decode here, with a warranty on both cards. Also 1,150 W of graphics card.

What it cannot do. Two 575 W cards need their own circuit and a room that can shed 1.4 kW. On 230 V mains that is manageable, which it would not be on 120 V.

HIGH TIER · OPTION H2 · NEW, WARRANTED

Two RTX PRO 5000 Blackwell - 96 GB ECC

DELIVERED AED 54,919	VRAM 96 GB	ALL-IN MONTHLY AED 2,415	PER M TOKENS 5.95	DESIGN POINT Serves 20 orgs, headroom
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Blackwell, so native FP8 and the halved KV cache that comes with it. ECC memory, blower coolers, two slots and 300 W each: cards designed to sit in a workstation that runs continuously, which is exactly what this machine is. Two reach 96 GB on a 1200 W supply where the PRO 6000 build needs more card budget and more cooling. The reason it is not the headline recommendation is the quote spread - Promise Gulf's product page says AED 15,500 and its own price list says AED 31,943 for the same part. Get the number in writing before planning around it.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX PRO 5000 Blackwell 48 GB ECC, new x2	GCC Gamers	AED 17,500	AED 35,000
AMD Ryzen Threadripper 7960X (24C, sTR5, 88 lanes)	GCC Gamers	AED 6,150	AED 6,150
Gigabyte TRX50 AERO D, sTR5, 4x x16	—	AED 2,600	AED 2,600
128 GB DDR5 RDIMM (TRX50, quad channel)	—	AED 6,000	AED 6,000
4 TB NVMe Gen4	Microless	AED 2,020	AED 2,020
Full-tower case + fans	—	AED 500	AED 500
sTR5 air cooler	—	AED 500	AED 500
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1200 W 80+ Gold/Platinum	Alershad	AED 899	AED 899
Retail subtotal			AED 54,769
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 54,919
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 1,050 failure reserve)			AED 53,354

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	96 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	yes, local warranty	✓

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 6000 96 GB, \$2.20/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 1,967	AED 2,218	AED 903	30 mo	26 mo	380 W
12 h/day	AED 2,951	AED 3,327	AED 933	19 mo	17 mo	450 W
24 h/day	AED 5,902	AED 6,653	AED 1,023	9 mo	8 mo	660 W

Verdict

The overlooked option. The same 96 GB and the same ECC as the PRO 6000 build, for around three quarters of the price and half the power.

What it cannot do. Two cards rather than one, so tensor parallelism has to be configured and a model larger than 48 GB pays a PCIe crossing. Local pricing on this part varies by more than 30%, which is a procurement risk in itself.

HIGH TIER · OPTION H3 · NEW, WARRANTED

One RTX PRO 6000 Blackwell - 96 GB ECC

DELIVERED AED 71,919	VRAM 96 GB	ALL-IN MONTHLY AED 2,879	PER M TOKENS 7.10	DESIGN POINT Serves 20 orgs, headroom
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96 GB of ECC memory on one card at 1,792 GB/s, 600 W, one slot, no tensor parallelism to configure and no second-hand risk. One card is one thing to monitor, cool and replace, and that operational simplicity is worth real money to a small team. The difficulty is what the market has done to it: it listed at USD 8,565 at launch and USD 16,000 now, and none of that increase bought performance. Locally the same part number ranges from AED 48,825 to AED 77,032 depending on who answers the phone, so the first task is a written quote rather than a decision.

Bill of materials, at UAE seller prices

Item	Seller	Unit	Line
RTX PRO 6000 Blackwell 96 GB ECC, new	GCC Gamers	AED 52,000	AED 52,000
AMD Ryzen Threadripper 7960X (24C, sTR5, 88 lanes)	GCC Gamers	AED 6,150	AED 6,150
Gigabyte TRX50 AERO D, sTR5, 4x x16	—	AED 2,600	AED 2,600
128 GB DDR5 RDIMM (TRX50, quad channel)	—	AED 6,000	AED 6,000
4 TB NVMe Gen4	Microless	AED 2,020	AED 2,020
Full-tower case + fans	—	AED 500	AED 500
sTR5 air cooler	—	AED 500	AED 500
1500 VA line-interactive UPS	—	AED 1,100	AED 1,100
1200 W 80+ Gold/Platinum	Alershad	AED 899	AED 899
Retail subtotal			AED 71,769
Delivered, VAT-inclusive (includes AED 150 local delivery)			AED 71,919
Effective capital cost (less 5% VAT, recoverable if registered, plus AED 1,560 failure reserve)			AED 70,054

Capability against the design point — 42 GB at 15 organisations, 43 GB at 20

Capability	This option	
Pooled memory vLLM can use	96 GB	✓
Continuous batching	yes, CUDA + vLLM	✓
Matched cards, tensor parallel	yes	✓
FP8 KV cache	yes	✓
Warranty and tax invoice	yes, local warranty	✓

Buy versus rent, at three duty cycles

Duty cycle	Rent abroad RTX PRO 6000 96 GB, \$2.20/hr	Rent in UAE H100 80 GB, GPU.ai UAE, \$2.48/hr	Own, monthly run	Break-even vs abroad	Break-even vs in region	Avg draw
8 h/day	AED 1,967	AED 2,218	AED 903	39 mo	34 mo	380 W
12 h/day	AED 2,951	AED 3,327	AED 933	25 mo	22 mo	450 W
24 h/day	AED 5,902	AED 6,653	AED 1,023	12 mo	11 mo	660 W

Verdict
Technically the best machine in this report, at the top of the budget and the top of a supply squeeze.

What it cannot do. Nothing technical. What it cannot do is stay busy: at the specified workload it would idle most of the day, and it consumes most of the AED 80,000 ceiling on its own.

Business analysis

Two baselines, one total cost of ownership, and the conditions under which the answer flips.

There is no revenue model here by design. This decision is judged on cost avoidance against two alternatives that already exist and can be priced today: paying an API for the tokens, and renting the same capability by the hour. Anything resting on a seat price nobody has set yet would be a forecast dressed as an analysis.

How a monthly cost is built

Component	Basis	Build L2
Capital, amortised	Effective capital over 36 months, VAT stripped, failure reserve added	AED 428/mo
Electricity	Average draw at the DEWA top commercial slab, AED 0.440/kWh including the fuel surcharge	AED 151/mo
Cooling	×1.33 on electricity — a split unit moves roughly three watts of heat per watt drawn	AED 50/mo
Operations	4 hours a month at AED 185/hr — patching, drivers, model swaps, the 3am restart	AED 740/mo

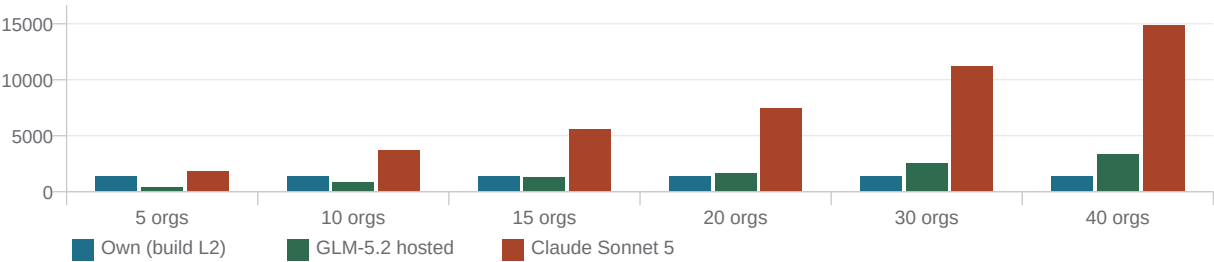
Total **AED 1,369/mo**

The same four components are applied identically to all nine options. Operations is charged to the rental options too, at the same rate: renting removes hardware maintenance but adds orchestration, and pretending otherwise would bias the comparison. On the cheaper builds it is the largest single line — below a certain hardware cost, the machine is not what you are paying for.

Baseline one — what the tokens would cost from an API

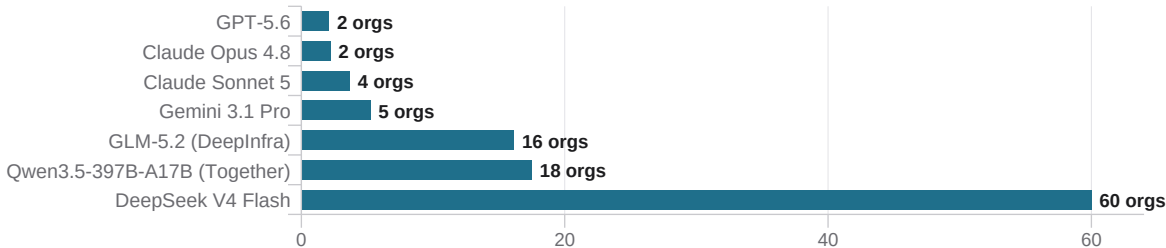
The design point generates **406 million tokens a month**. The ratio matters as much as the total: at 6,000 prompt tokens to 400 output, this workload is about 94% input, which makes input pricing dominant and blends the headline rates downward. That flatters the APIs rather than the reverse.

Model	Class	Blended \$/M	Blended AED/M	Monthly, AED	vs L2
GPT-5.6	frontier	6.562	24.10	AED 9,773	7.1×
Claude Opus 4.8	frontier	6.250	22.95	AED 9,308	6.8×
Claude Sonnet 5	frontier	3.750	13.77	AED 5,585	4.1×
Gemini 3.1 Pro	frontier	2.625	9.64	AED 3,909	2.9×
GLM-5.2 (DeepInfra)	open-hosted	0.853	3.13	AED 1,270	0.9×
Qwen3.5-397B-A17B (Together)	open-hosted	0.787	2.89	AED 1,173	0.9×
DeepSeek V4 Flash	open-hosted	0.118	0.43	AED 175	0.1×
Build L2, owned	on-prem	—	3.38	AED 1,369	—



Monthly cost in AED against client count. The owned line is flat because capital, power and operations do not care how many tokens pass through. That flatness is the entire economic argument for owning, and it is why the case strengthens with every client added.

Where the lines cross



Client organisations at which owning build L2 becomes cheaper than paying per token. Bars are capped at 60. Against a frontier API the machine pays for itself almost immediately; against a hosted open-weight model it needs the top of the target range.

The competitor is not the one you would expect

The instinct is to compare an owned machine against GPT or Claude, where it wins easily — Claude Sonnet 5 would cost AED 5,585 a month against AED 1,369 owned. **The real competitor is a hosted open-weight model at AED 3.13 per million tokens**, running the same class of model this machine would run, for AED 1,270 a month. On cost per token alone that comparison is only won above about 16 client organisations. The justification for owning has to rest on the three things a hosted option structurally cannot offer: **documents that never leave the building, latency with no data-centre round trip, and a bill that does not move when usage does.**

Baseline two — buy versus rent, twice

Because there is no in-region rental of a 48 GB consumer card, the two rental lines are not the same product. Internationally the design point rents for \$0.64 an hour. The cheapest UAE-hosted option is a datacentre H100 at \$2.48 — larger hardware than this workload needs, at about four times the price. Both are shown, because the difference between them is the price of a compliance position rather than a price of compute.

Break-even in months, against **renting abroad** (first three columns) and **renting in the UAE** (last three)

Option	Effective capital	8 h/day	12 h/day	24/7	8 h/day	12 h/day	24/7
L2 Two used RTX 3090 - 48 GB	AED 15,404	38	23	11	8	5	2
L3 One new RTX 5080 - 16 GB †	AED 15,305	83	49	22	7	5	2
M1 Two used RTX 4090 - 48 GB	AED 30,631	82	50	23	15	10	5
M2 One new RTX 5090 - 32 GB †	AED 24,992	167	93	40	12	8	4
M3 Four used RTX 3090 - 96 GB	AED 30,761	19	12	6	16	10	5
H1 Two new RTX 5090 - 64 GB	AED 49,937	29	19	9	26	17	8
H2 Two RTX PRO 5000 Blackwell - 96 GB ECC	AED 53,354	30	19	9	26	17	8
H3 One RTX PRO 6000 Blackwell - 96 GB ECC	AED 70,054	39	25	12	34	22	11

Green is a break-even inside the 36-month write-off window. † marks an option that does not meet the design point, so a fast payback there is buying something other than the specified workload. Note the pattern: against international rental most builds struggle, and against in-region rental almost all of them repay quickly. That single contrast is the decision.

The inversion, stated plainly

At twelve hours of GPU duty a day, renting a capable card abroad costs AED 858 a month and owning build L2 costs AED 629 in capital and power — close enough that the choice turns on preference rather than arithmetic. Renting the same capability in the UAE costs AED 3,327 a month, roughly 5 times owning. **If the data may leave the country, rent. If it may not, buy.** There is no third reading of these numbers.

Three-year total cost of ownership

Option	Capital	Power + cooling	Operations	3-year TCO	AED/M tok
L1 Rent, own nothing	AED 30,905 rent	—	AED 26,640	AED 57,545	3.94
L2 Two used RTX 3090 - 48 GB	AED 15,404	AED 7,244	AED 26,640	AED 49,288	3.38
L3 One new RTX 5080 - 16 GB	AED 15,305	AED 4,624	AED 26,640	AED 46,569	3.19
M1 Two used RTX 4090 - 48 GB	AED 30,631	AED 8,786	AED 26,640	AED 66,057	4.53
M2 One new RTX 5090 - 32 GB	AED 24,992	AED 6,281	AED 26,640	AED 57,913	3.97
M3 Four used RTX 3090 - 96 GB	AED 30,761	AED 13,564	AED 26,640	AED 70,965	4.86
H1 Two new RTX 5090 - 64 GB	AED 49,937	AED 11,637	AED 26,640	AED 88,215	6.04
H2 Two RTX PRO 5000 Blackwell - 96 GB ECC	AED 53,354	AED 6,936	AED 26,640	AED 86,930	5.95
H3 One RTX PRO 6000 Blackwell - 96 GB ECC	AED 70,054	AED 6,936	AED 26,640	AED 103,630	7.10

Three years at twelve hours of daily GPU duty, international rental line. Operations is a flat AED 26,640 on every row.

Sensitivity: what survives being wrong

Scenario	Change	AED/M	Crossover	Consequence
Base case	As modelled	3.38	16 orgs	The recommendation stands.
Half the utilisation	Token volume -50%	6.75	16 orgs	Cost per token doubles; the API comparison worsens, the rental comparison does not move. Strengthens the case for renting.
Triple the volume	Token volume ×3	1.13	—	Owning beats every option on this page including the cheap hosts. This is the scenario that justifies buying, and it is measurable in advance.
32k context sessions	KV cache doubles	3.38	unchanged	Cost per token is unchanged but 48 GB stops fitting. This changes the <i>option</i> , not the economics: L2 and M1 become under-sized and H2 becomes the answer.
Card prices fall 30%	Memory market normalises	3.06	15 orgs	Break-evens shorten by about a third and the high tier becomes defensible. The reason the 96 GB options are deferred rather than dismissed.
Operator time doubles	8 hours a month	5.20	25 orgs	The most under-estimated line in any on-premise case, and on the cheap builds it already dominates the hardware.

What survives all six

Two conclusions hold in every scenario. **The vLLM work comes first** — it is free, and no scenario makes serial inference acceptable for ten clients. **Renting briefly to measure before committing capital is never the wrong move** — where owning wins, a month of rental delays it by a month and costs AED 1,692; where owning loses, it saves between AED 15,569 and AED 71,919.

Depreciation and residual value

Standard practice writes compute hardware down to nothing over three years and this report's TCO does exactly that. The current market makes that conservative in one specific place: second-hand Ampere has held its price while everything new inflated around it, so cards bought used today are unlikely to be worth nothing in three years. The downside of the recommended build is smaller than the amortisation table implies.

The mirror image is less comfortable. A card bought new at the top of a supply squeeze carries genuine mark-to-market risk: if memory supply normalises, the card does not become slower, but its resale value tracks a replacement cost that has fallen. **Buying used silicon at a market price is a smaller bet than buying new silicon at a shortage price**, and that asymmetry is a legitimate input to the decision.

Risk register

Risk	Likelihood	Impact	Mitigation
Residency answer arrives after a purchase	Medium	High	The whole decision hinges on it and it costs nothing to ask. Get it in writing before any card is bought; the wrong order here wastes the entire capital budget.
Used card fails, no warranty, no invoice	Medium	Medium	12% of card cost reserved in every second-hand build. Inspect in person, ask for power-on hours under roughly 20,000, run a sustained load test before paying, and meet somewhere the card can be plugged in. A failed 3090 is a AED 2,400 problem.
Quoted price differs from the listed price	High	Medium	UAE listings on the same part number span more than 50% on the workstation cards. Treat every figure here as a starting point for a written quote, not a commitment.
Single box is a single point of failure	Medium	High	The real exposure once clients are contractual. Mitigation is a documented rental fallback that the vLLM port makes possible: an OpenAI-compatible client can be re-pointed at a rented endpoint in a configuration change. Do not sign an availability commitment before that path is tested.
Windows to Linux migration slips	Low	Medium	The systemd unit exists, paths go through pathlib, settings live in .env. Rehearse it on the rented instance in step 3, before any hardware is bought.
Model licence changes under a client	Low	High	Pin the exact revision, keep the LICENSE file with the deployment record, and prefer Apache 2.0 and MIT throughout — which the recommended stack already is.
Concurrency is far above the model	Medium	Medium	The reason step 3 is a month of measurement. Being wrong is only expensive if hardware is bought first.

Recommendation

Four stages, each with the number that triggers the next.

The cheapest mistake available is renting for a month and discovering the load is smaller than modelled. The most expensive is committing most of the budget to silicon at the top of a shortage, for a workload nobody has measured, to satisfy a compliance requirement nobody has confirmed. The plan below is ordered so each stage produces the evidence the next one needs, and so stopping after any stage leaves a working system.

STAGE 1 free · do this first

Ask whether the data may leave the country

Every number in this report bends around the answer. If client documents may be processed abroad, renting at AED 1,598 a month is competitive with owning and infinitely more reversible, and a hosted open-weight API at AED 1,270 is cheaper than both. If they may not, in-region rental costs AED 4,067 a month and owning becomes decisively correct.

Gate: a written answer from the client. **Nothing else on this page should start before it exists**, because it changes which half of the report applies.

STAGE 2 ~1 day · free, and required either way

Lift the three ceilings

Port `app/llm.py` from Ollama's `/api/chat` to an OpenAI-compatible `/v1/chat/completions`. Raise **JOB_WORKERS** once a second lane exists. Run uvicorn with more than one worker. This is needed whether the endpoint ends up rented or owned, and it is what lets a rented endpoint act as a fallback later.

Gate: the full suite stays green against a vLLM endpoint — `pytest`, `check_tools`, `check_agent`, `check_search`, `check_database`, `check_endtoend`.

STAGE 3 ~AED 1,692 · reversible

Rent for a month and measure

A 48 GB instance running the real stack against real client documents on Ubuntu. It replaces the modelled concurrency figure with a measured one and rehearses the Linux migration before any card is bought. Benchmark SGLang against vLLM here, and settle quantisation with the project's own gates rather than a leaderboard. If residency was confirmed in stage 1, run this on the in-region line and accept the higher rate for the month.

Gate: record peak concurrent in-flight requests, p95 turn latency, tokens per session and prefix-cache hit rate. **Trigger to proceed:** measured peak concurrency at or above 6, sustained over several working days.

STAGE 4 AED 15,569 · the recommended purchase

Buy L2 — two used RTX 3090, 48 GB

Two matched 24 GB cards from the local second-hand market, on an AM5 platform with a 1200 W supply. It is the cheapest hardware that meets the design point and it leaves the most options open: if one card dies it is a AED 2,400 replacement, not a reconsidered decision. Step up to M1 — two used RTX 4090 at AED 29,769 — only if the twenty-organisation case is firm and the FP8 headroom is genuinely wanted.

Do not proceed if stage 1 said the data may leave the country and stage 3 measured peak concurrency below four. At that load renting is cheaper and the capital is better spent on the structured field extraction `HANDOVER.md` already names as the highest-value unbuilt thing.

What not to buy, and why

Option	Why not
L3 — new RTX 5080 16 GB	At the same money as two used 3090s it carries a third of the memory and cannot serve the workload. Worth having as a development box beside a rented endpoint; not as the server.
M2 — new RTX 5090 32 GB	32 GB does not hold the agent, a vision model and eight sessions of cache. The fastest single card here and the wrong size, at AED 15,000 for the privilege.
M3 — four used RTX 3090	96 GB of Ampere with no FP8 path, on an sTR5 platform that costs more than the cards. Four unwarranted cards solving a problem two already solve.
H3 — RTX PRO 6000 96 GB	Technically the best machine in this report. It also consumes 90% of the budget ceiling for capacity the measured workload does not need, at a price that rose 87% in eighteen months without the silicon changing. If a 96 GB machine is truly wanted, H2 reaches the same memory for AED 17,000 less.

The one thing to settle before any of it

Confirm the residency requirement in writing

It is worth repeating because it is both the cheapest step and the one most likely to be skipped. If documents may leave the UAE: renting abroad at AED 1,598 a month is competitive with owning, and a hosted open-weight API at AED 1,270 is cheaper than every option in this report below about 16 client organisations — in which case the capital is better spent on the product than on the infrastructure. If they may not: in-region rental at AED 4,067 a month makes owning repay itself in well under a year, and build L2 at AED 15,569 is the right purchase. **Two different reports follow from one question, and it costs an email to ask.**

Appendix A — what to measure before spending

Every concurrency figure in this report is modelled. Four instruments turn them into measurements.

Metric	Where it goes	Why it decides something
Peak concurrent in-flight requests	A gauge around the agent call in app/agent.py , sampled per second and reported on /api/health	The single input the whole VRAM budget hangs on. Section 3 models 6 at fifteen organisations; if the real figure is three, buy nothing.
p95 turn latency, split prefill / decode	Timestamps already available in the steps trace agent.ask() returns	Separates 'the model is slow' from 'the tools are slow'. If tool round-trips dominate, a faster card changes nothing.
Tokens per session, prompt and output separately	Log the counts the inference server already returns	Drives both the KV budget and the API comparison. The 94% input ratio is an assumption; measuring it could move the blended rates materially.
Prefix-cache hit rate	vLLM exposes it directly once stage 1 lands	On an agent loop that re-sends its whole history, this is the cheapest performance lever available and nobody currently knows its value here.

All four are cheap to add and none requires new infrastructure. The project already has the right habit — `check_services.py` compares a fingerprint of running code against what is on disk rather than trusting a restart — and this is the same instinct applied to load.

Appendix B — assumptions

Assumption	Value	Basis
Active users per organisation	4	Assumed
Sessions per user per day	12	Assumed
Turns per session	4	Assumed
Prompt tokens per turn	6,000	Derived from <code>TOOL_RESULT_BUDGET = 12,000</code> chars and the full-history re-send in <code>app/agent.py</code>
Output tokens per turn	400	Assumed
Peak hour share of daily traffic	20%	Assumed; deliberately pessimistic for an office-hours shape
Context per session	16,384	Assumed; qwen3 models here are trained to 40,960
Target decode rate per user	25 tok/s	The rate at which streamed output reads as responsive
Prefill rate share under batching	2,500 tok/s	Modelled. The measured <code>prompt_tps</code> in <code>bench_results.json</code> is not usable for this: it reports cached prompt processing
Agent KV geometry	48 layers, 4 KV heads, 128 head dim	Assumed for a 30B-A3B-class model. The formula is validated against the measured 8B, not against this configuration
vLLM memory utilisation	0.90	Practical default; you do not get the whole card
Amortisation	36 months	Standard for compute hardware
Operator time	4 h/mo at AED 185/hr	Mid-market UAE contract rate; charged identically to buying and to renting
Used-card failure reserve	12% of card cost	No warranty on second-hand silicon
Electricity	AED 0.440/kWh	DEWA top commercial slab 0.380 AED plus the 0.060 AED fuel surcharge. A server running continuously sits in the top slab, so this is the right marginal rate
Cooling uplift	×1.33	A split unit moves roughly three watts of heat per watt drawn
Purchase basis	UAE seller price less 5% VAT, plus AED 150 delivery	Bought locally, so customs duty is already inside the shelf price and there is no freight or clearance. VAT is recoverable by a registered business. USD-billed APIs and international rental convert at the pegged AED 3.6725, so there is no currency risk

Appendix C — sources

Every figure in this report resolves to a row in `docs/hardware_report_data.json` carrying the URL and capture date below. The generator refuses to build if a priced row is missing either.

Captured	Source
2026-09-03	https://getdeploying.com/gpus/nvidia-a100
2026-09-02	https://getdeploying.com/gpus/nvidia-rtx-5090
2026-09-03	https://getdeploying.com/gpus/nvidia-rtx-pro-6000
2026-09	https://benchlm.ai/llm-pricing
2026-09	https://costtocost.ae/ram-memory/
2026-09	https://gccgamers.com/computer-parts-compnents/amd-ryzen-9-9950x-16-cores-32-threads-desktop-processor-100-100001277wof.html
2026-09	https://gccgamers.com/computer-parts-compnents/amd-ryzen-threadripper-7960x-24cores-48threads-4-2-ghz-processor-100-100001352wof.html
2026-09	https://gccgamers.com/computer-parts-compnents/graphic-cards/rtx-5090.html
2026-09	https://gccgamers.com/computer-parts-compnents/nvidia-rtx-pro-5000-blackwell-48gb-gddr7-ecc-graphics-card.html
2026-09	https://gccgamers.com/top-selling-products/pny-nvidia-rtx-pro-6000-96gb-blackwell-workstation-edition-oem-graphics-card-dlss-4.html
2026-09	https://getdeploying.com/gpus/nvidia-rtx-pro-5000
2026-09	https://uae.dubizzle.com/classified/computers-networking/computer-components/graphics-video-cards/
2026-09	https://uae.microless.com/graphic_cards/rtx-4090/326//
2026-09	https://uae.microless.com/motherboards/am5/340//
2026-09	https://uae.microless.com/ssd/m-2/329//
2026-07	https://utilitybilluae.com/business-electricity-tariffs-uae/
2026-09	https://www.alershadonline.com/collections/1200w-1650w-power-supplies
2026-09	https://www.amazon.ae/PNY-GeForce-RTXTM-Triple-Graphics/dp/B0DTJDR3V9
2026-08	https://www.tomshardware.com/pc-components/gpus/nvidia-doubles-rtx-pro-6000-blackwells-msrp-to-a-staggering-usd16-000-96gb-card-started-pre-orders-below-usd8-000-last-year
2026	https://docs.vllm.ai/en/latest/getting_started/installation.html
2026	https://gigagpu.com/vllm-on-rtx-5090-max-throughput/
2026	https://intuitionlabs.ai/articles/h100-rental-prices-cloud-comparison
2026	https://markaicode.com/benchmarks/home-ai-server-benchmark/
2026	https://markaicode.com/benchmarks/rtx-5090-ai-benchmark/
2026	https://markaicode.com/benchmarks/rtx-5090-inference-benchmark/
2026	https://nodes.edgevana.com/gpu-servers/h100/uae
2026	https://tech-insider.org/gpu-prices-2026/
2026	https://www.astutegroup.com/news/general/gpu-pricing-set-for-reset-as-ai-driven-memory-shortages-push-costs-sharply-higher/
2026	https://www.cleartax.com/ae/vat-charges-on-electronics-in-dubai
2026	https://www.comgateway.com/blogs/is-an-rtx-5090-founders-edition-worth-aed-12-000-when-the-us-msrp-is-1-999/
2026	https://www.morphllm.com/comparisons/together-vs-deepinfra
2026	https://www.morphllm.com/llm-api-pricing
2026	https://www.spheron.network/blog/gpu-cloud-middle-east-2026/
2026	https://www.spheron.network/blog/llm-api-pricing-comparison-gpt-claude-gemini-deepseek-2026/

Internal sources — this project's own `bench_results.json`, `check_env.py` output, `README`, `HANDOVER.md` and `docs/plans` — are cited inline where used and are not listed here.

Appendix D — regenerating this report

GPU and memory prices in this market have moved by twenty per cent inside two months. This document is built to be re-run rather than re-written:

`python scripts/build_hardware_report.py`

Edit `docs/hardware_report_data.json` — a price, a rental rate, an API tariff, a workload assumption — and run the generator. The sizing model, every cost table, every break-even, every chart and the source appendix all re-derive from those inputs. Nothing in the prose contradicts the data file, because nothing in the prose is typed twice.

The script imports nothing from `app/`, so it cannot affect the running service, and it needs only `reportlab`, which is already in `requirements.txt`.

One habit worth keeping when it is re-run: record the capture date on every row at the moment the figure is found, not afterwards. Figures assembled after the fact are not wrong when they were written — they were assembled from working sessions whose own dates had already drifted past the market. A date on every number is what makes the difference visible next time.